Decision Trees

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

Before you start

Week 3: ordered questions route a case to one outcome.

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Fit a root question and leaf predictions from training examples. Try the recall before opening notes.

Your learning target

I can fit a **root** question and **leaf prediction**s from **training** examples.

I can trace a new example through the learned tree.

I can compare a shallow tree with a baseline and state its limits.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Types of Classification Error. Give an example and a mistake that the idea helps you avoid.

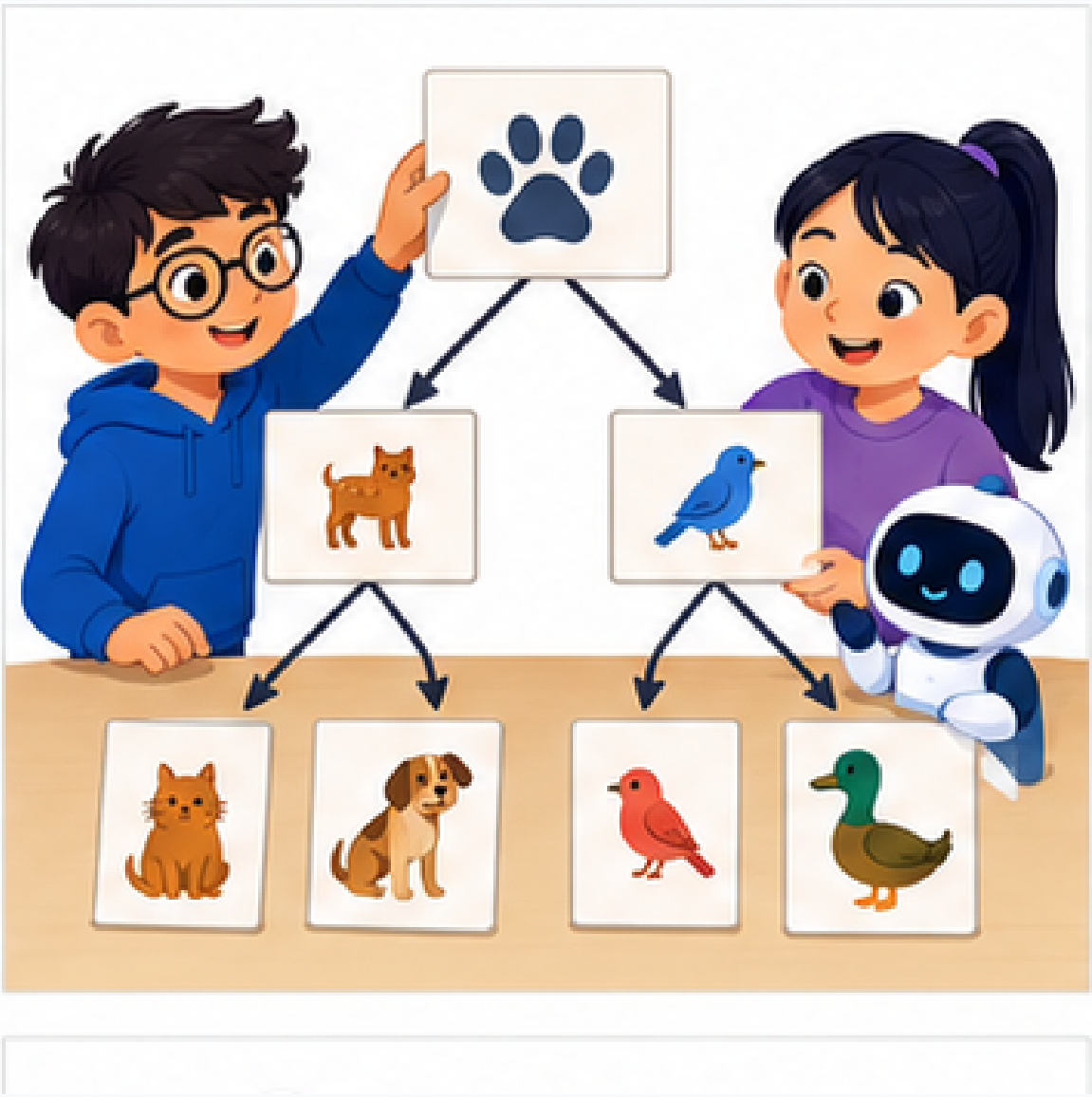
Without opening the old lesson, explain one key idea from **Training** and Testing. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Operational Labels. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 14: **Decision Tree**s

Each question **partition**s possibilities; question order affects efficiency.



THINKER MISSION

Build a tree that identifies eight mystery animals.

Thinker name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

> Why does a cutoff with zero **training** errors still need validation on separate examples?

This week's end goal

By the end, I can fit a one-question tree using **training** labels and an explicit candidate comparison.

STUDENT LESSON · 2 OF 18

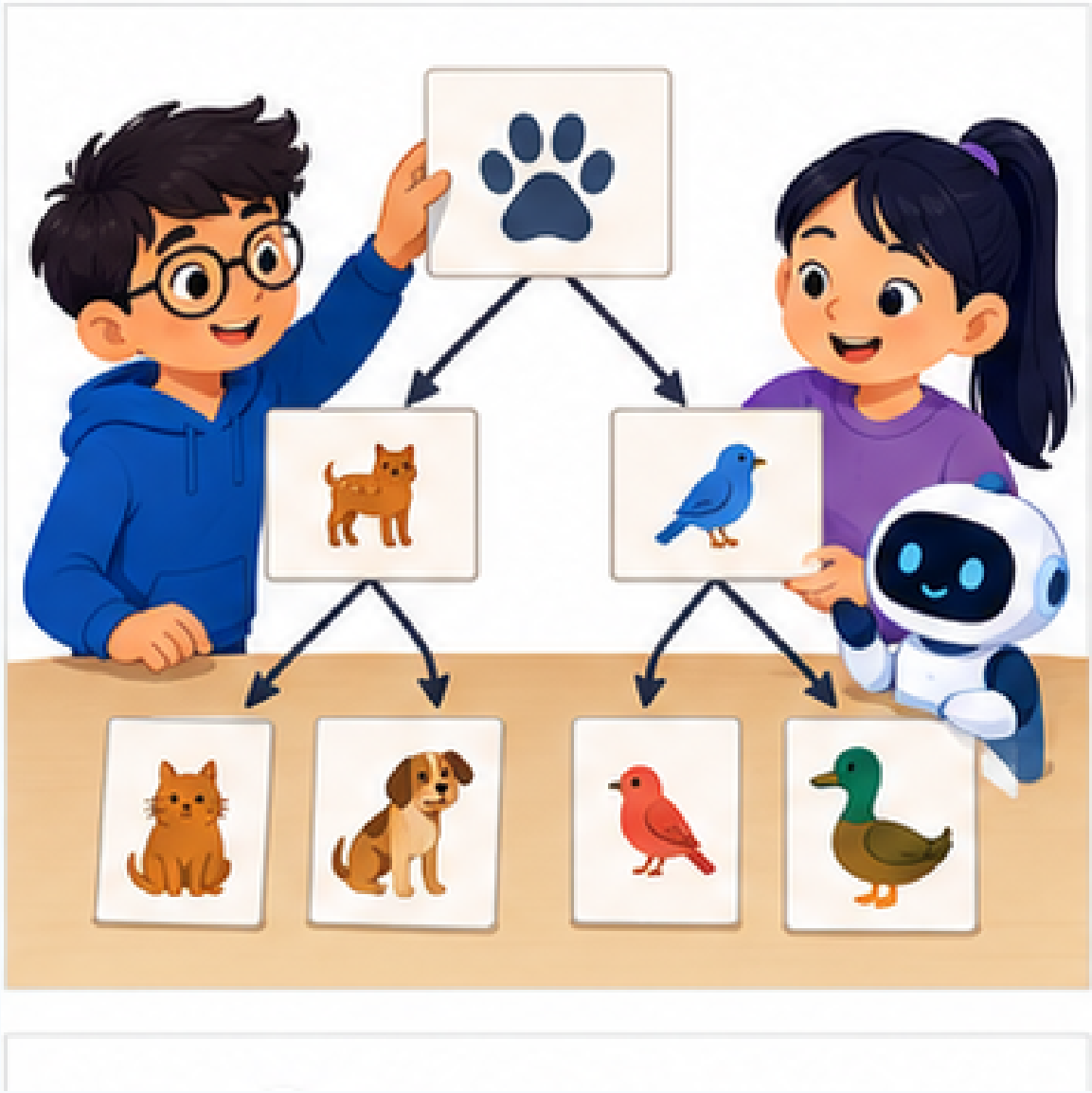
Notice the evidence

> Investigate the supplied Decision Trees case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use **decision tree**s to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

Write or draw your reasoning and evidence.

My **prediction** and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

A **decision tree** asks a feature question and follows a **branch** to a **prediction**. A tree with one question has **depth** one and is sometimes called a stump.

Unlike Week 8, where the two **output** labels were fixed in advance, this week we fit both the question and the label stored at each **leaf**.

Our six fictional **training** rows have two binary sensor flags available before **prediction**: near and tall. A value of 1 means that flag is present; 0 means absent. The reviewed scene label is separate.

Each row represents a different trip on a familiar route, including rows with identical values. Similar readings can have different scene labels. These are teaching examples, not robot control instructions.

Row	Near	Tall	Label
1	1	0	blocked
2	1	1	blocked
3	1	0	blocked
4	0	1	clear

Row	Near	Tall	Label
5	0	0	clear
6	0	1	blocked

Try near first and tall second. For each question, group the **training** rows by 0 and 1, then predict each group's majority label; choose clear for tied label counts.

Count all **training** mistakes and keep the first question if scores tie. Stop after one question. This is our stated classroom **fitting** procedure.

STUDENT LESSON · 3 OF 18

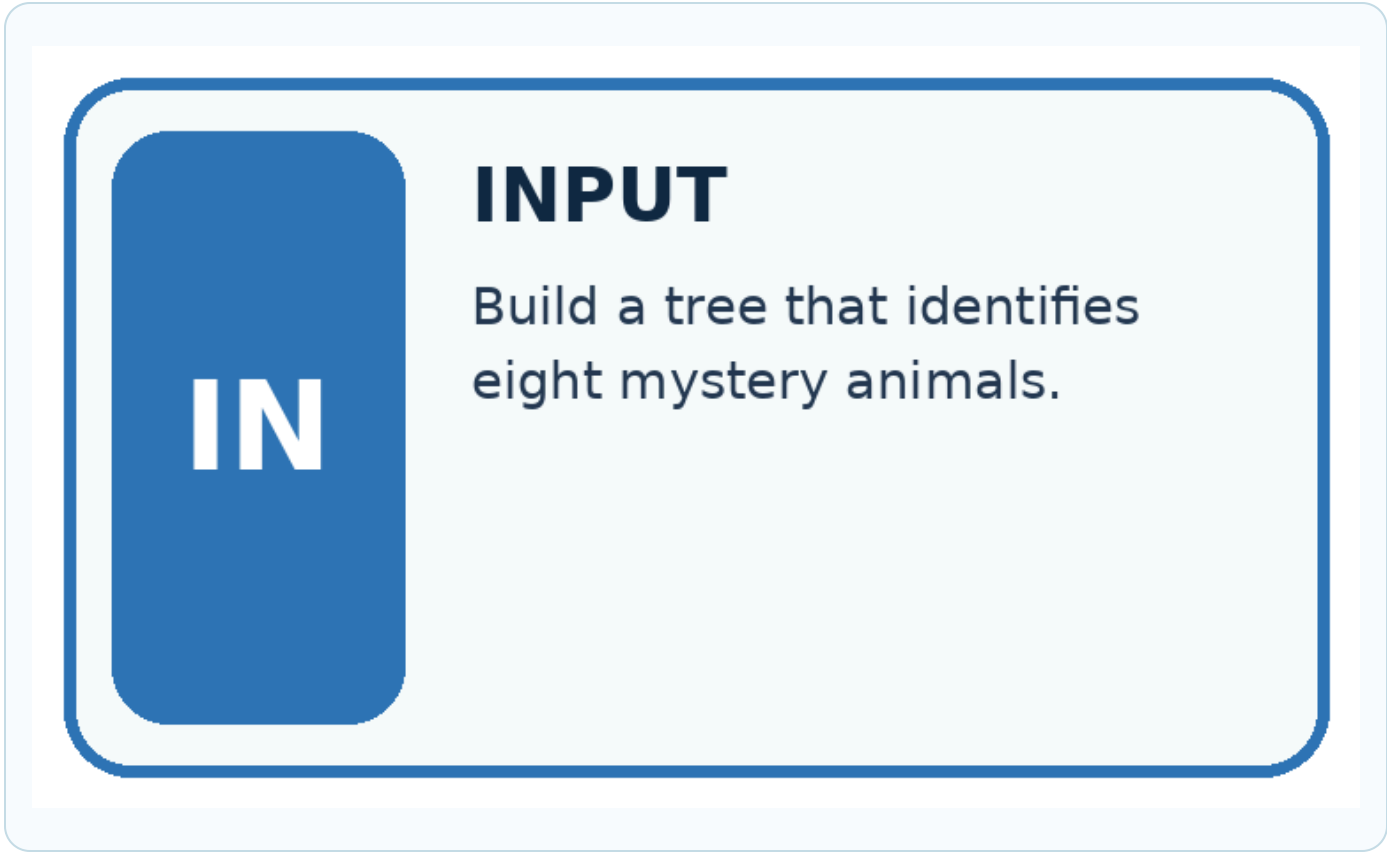
See how it works

> Follow the mechanism for Decision Trees. Connect each labelled input, deciding step and result.

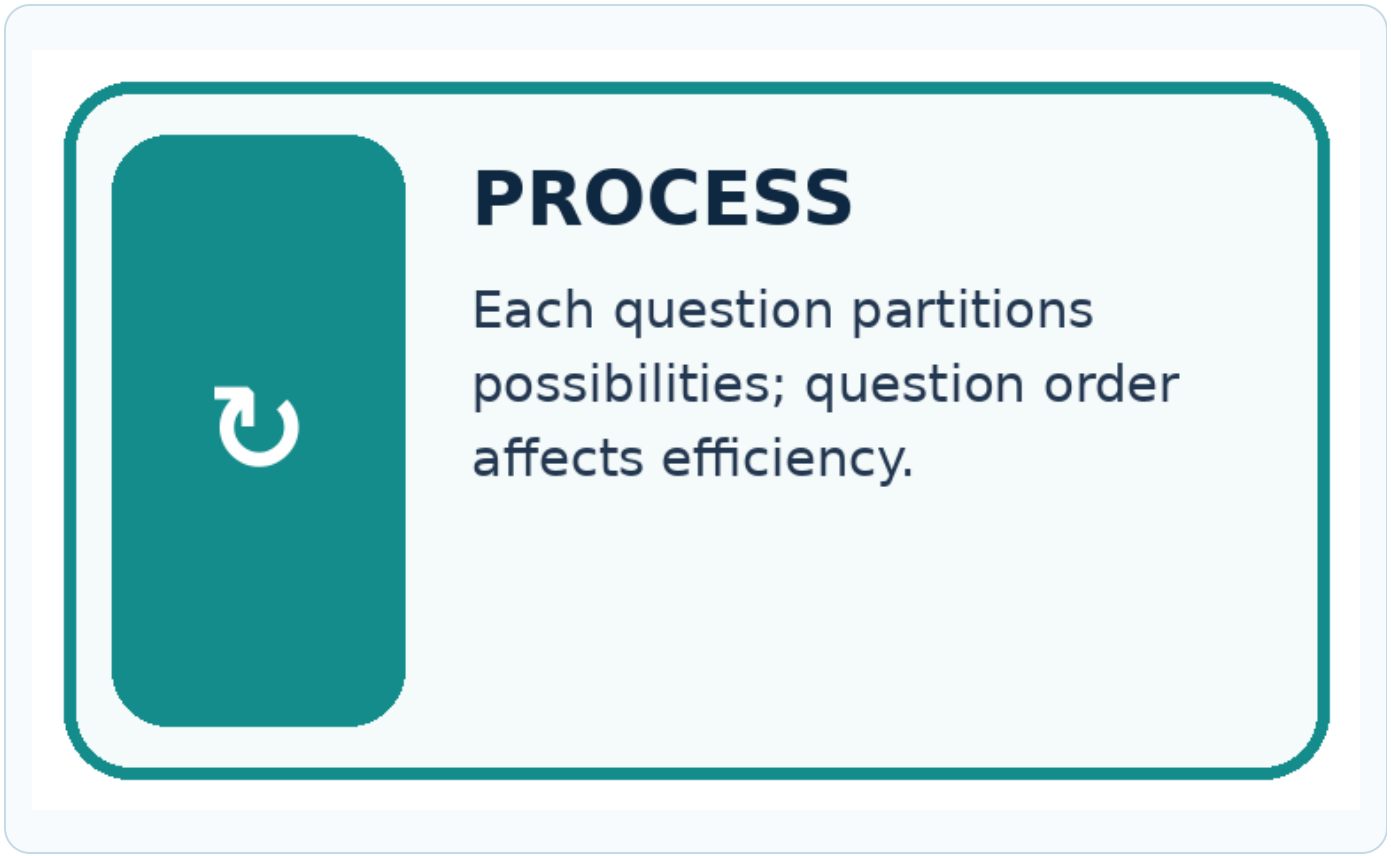
VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Build a tree that identifies eight mystery animals.

↓


PROCESS
Each question partitions possibilities; question order affects efficiency.

↓



Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

```
rows = [(1, 0, "blocked"), (1, 1, "blocked"),
        (1, 0, "blocked"), (0, 1, "clear"),
        (0, 0, "clear"), (0, 1, "blocked")]
def majority(labels):
    if labels.count("blocked") > labels.count("clear"):
        return "blocked"
    return "clear"
best_errors = len(rows) + 1
```

```
for feature in [0, 1]:
    leaves = {}
    for value in [0, 1]:
        labels = [row[2] for row in rows
                  if row[feature] == value]
        leaves[value] = majority(labels)
    errors = 0
    for row in rows:
        if leaves[row[feature]] != row[2]:
            errors = errors + 1
    print(feature, errors)
    if errors < best_errors:
        best_errors = errors
        tree = (feature, leaves)
print(tree[0], tree[1][0], tree[1][1])
```

Feature 0 is near; feature 1 is tall; row[2] is the label. **Output** is 0 1, then 1 2, then 0 clear blocked. Near makes one **training** mistake: near=0 contains two clear labels and one blocked, so that **leaf** predicts clear.

Near=1 contains three blocked labels. Tall makes two mistakes because both its groups have a blocked majority.

The stored tree contains the chosen feature and its two learned leaves. The code assumes nonempty data, binary flags, allowed labels and examples on both **branch**es of each candidate.

An empty **branch** would need a declared fallback; this small example does not implement one. The score is total **training** mistakes, not an average giving small and large **branch**es equal weight.

This is an explicitly limited **depth**-one learner. General tree algorithms can repeat split selection within **branch**es and often use impurity criteria such as Gini. Our code does not implement that full procedure.

Keeping the scope small lets you inspect every decision.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

Compare the cases

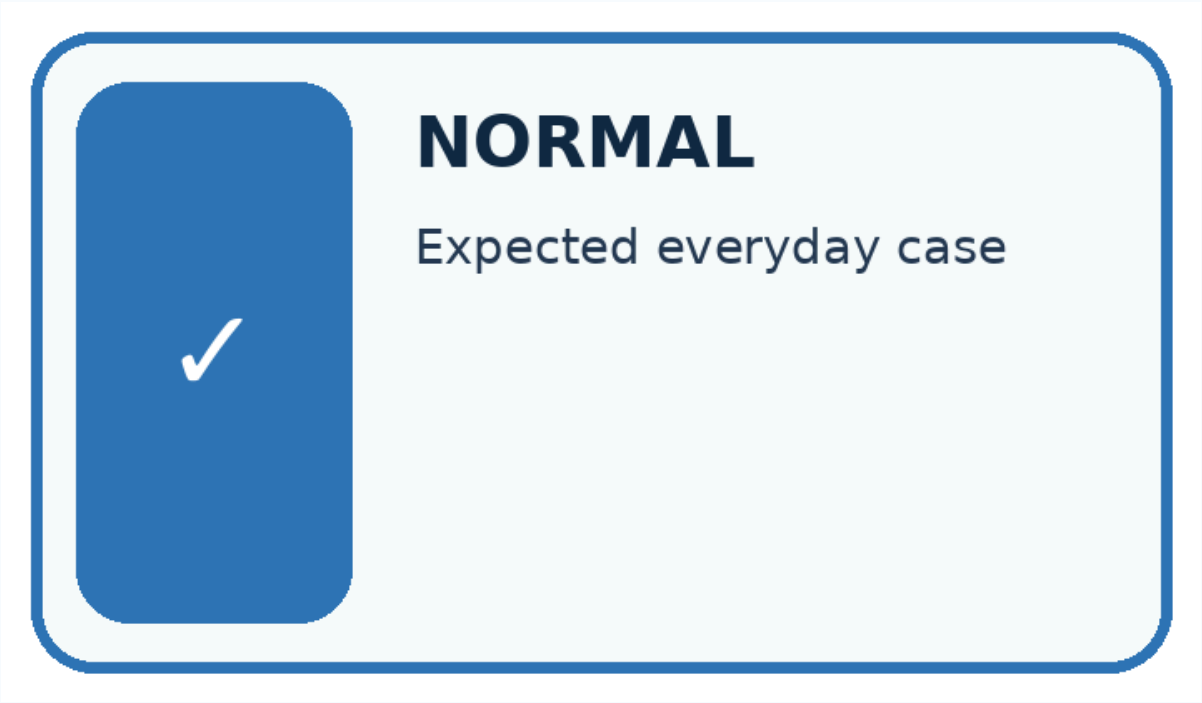
> Compare the supplied cases for Decision Trees. Identify the detail that changes the conclusion.

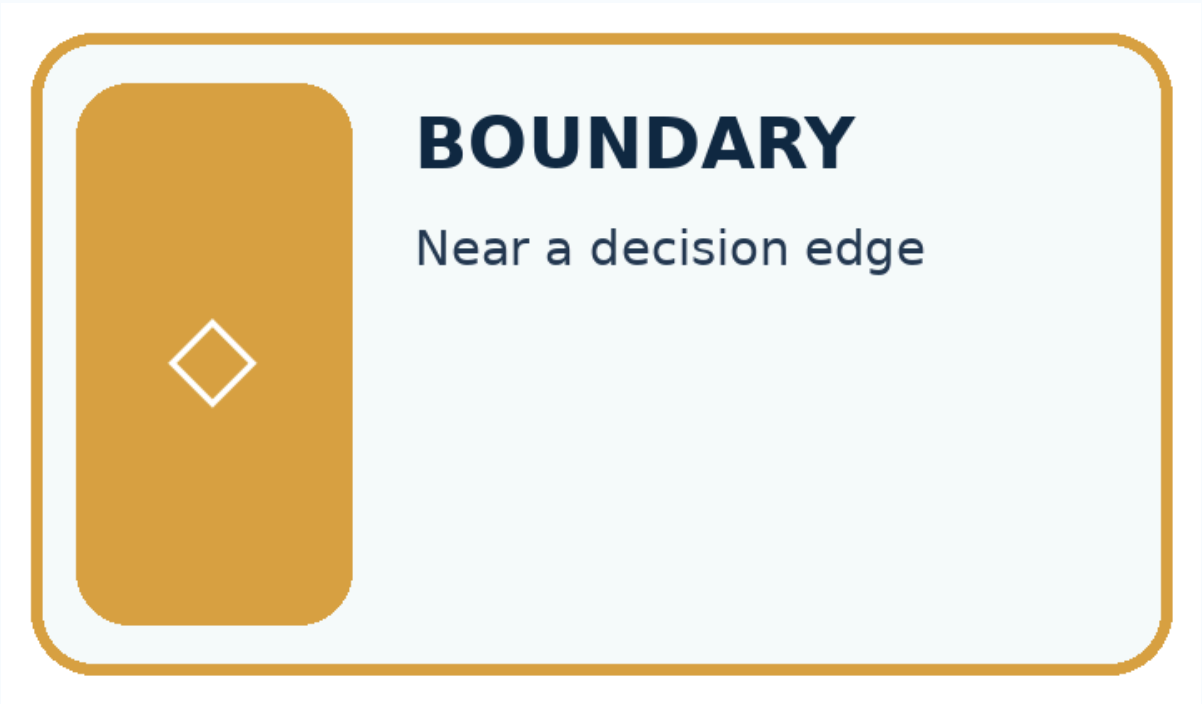
The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

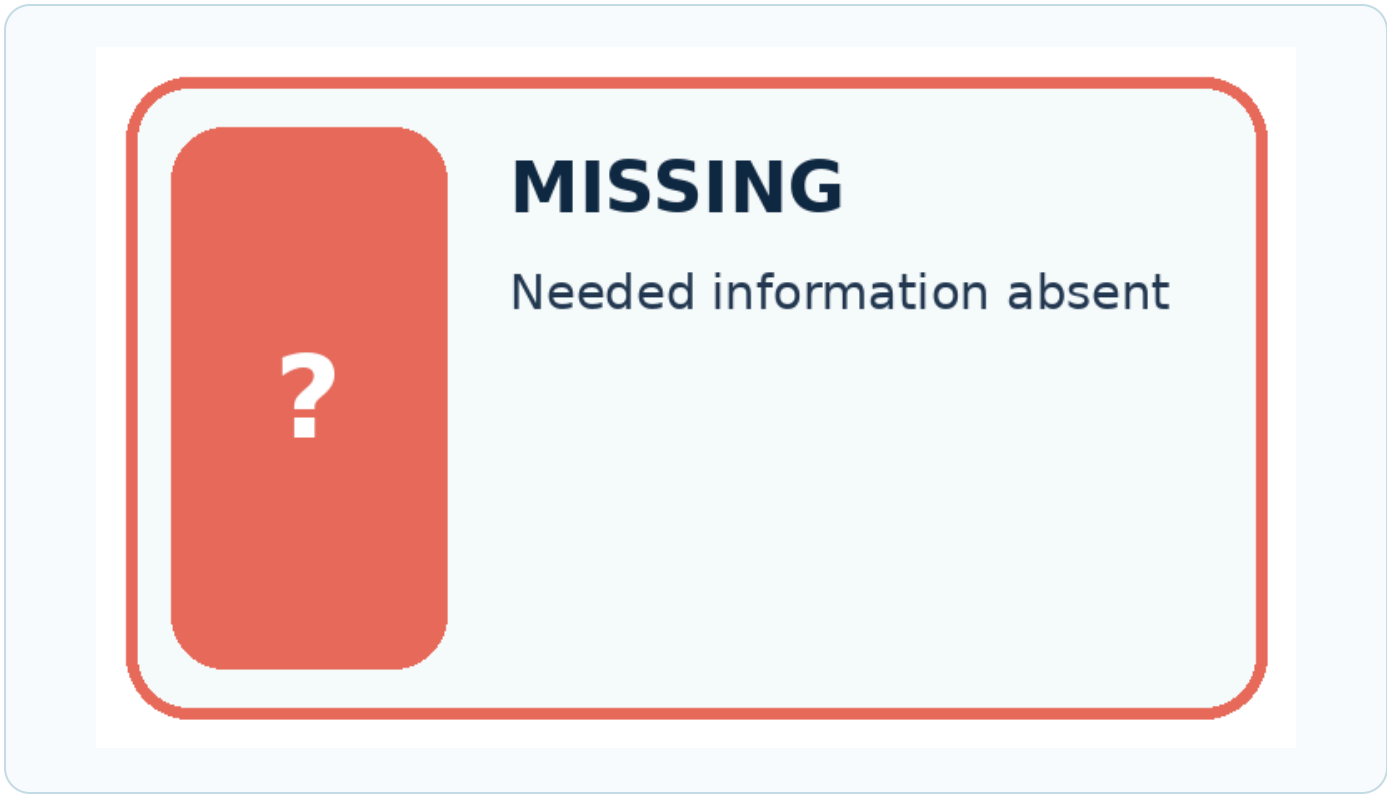
A system is not understood until it survives more than one comfortable example.





The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.


?
MISSING
Needed information absent


→
TRANSFER
Fresh situation or user

Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: New **input** becomes (0,1).

Predicted result

Prediction: The fitted tree predicts Yes.

Why the result follows

Reason: It follows diagram=1; cover colour is not the selected question.

Words that unlock the idea

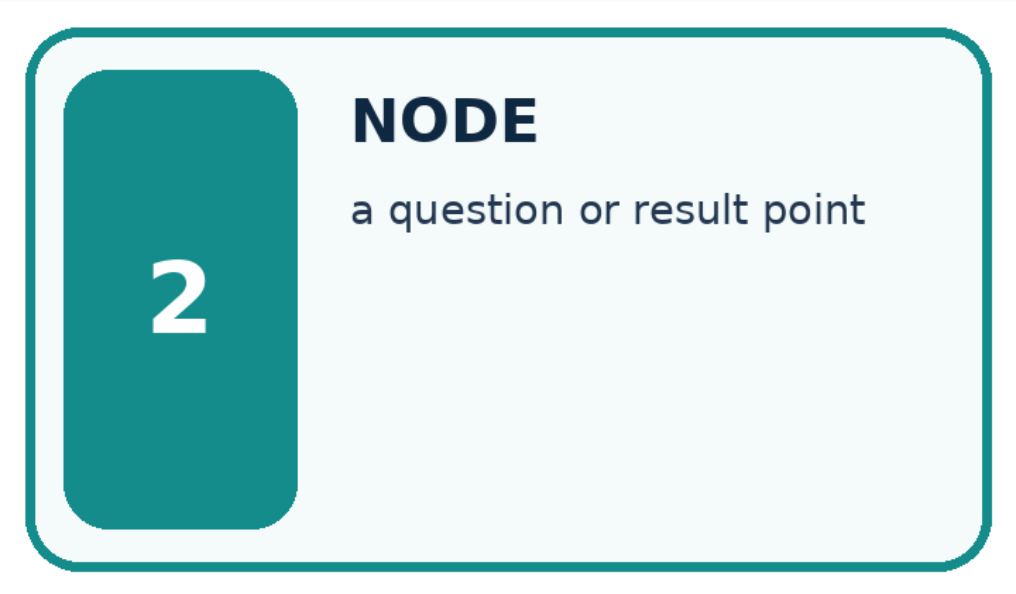
> Use the vocabulary of Decision Trees to explain a concrete decision, rather than only memorising definitions.

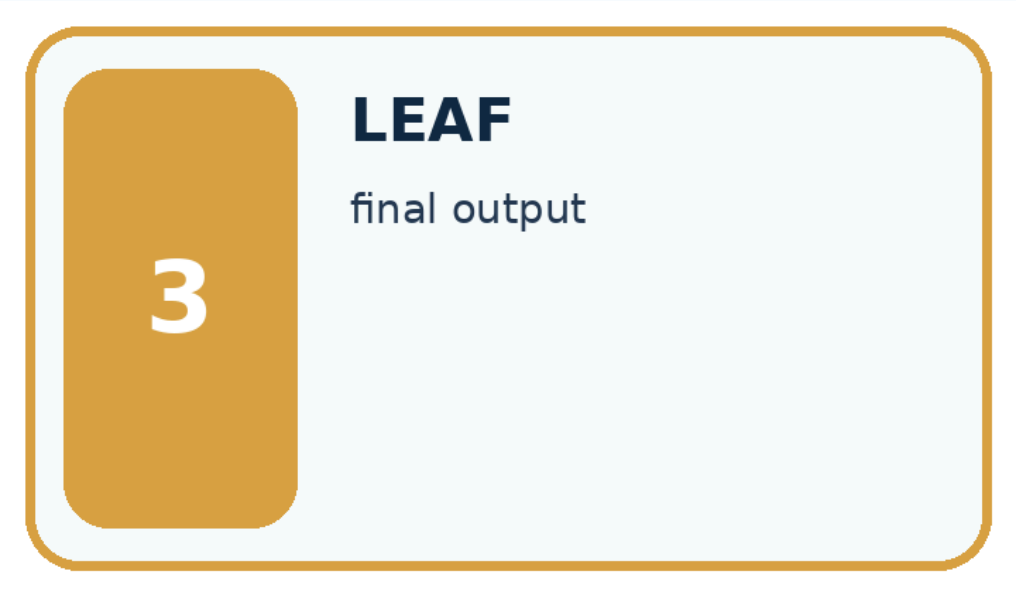
VOCABULARY LAB

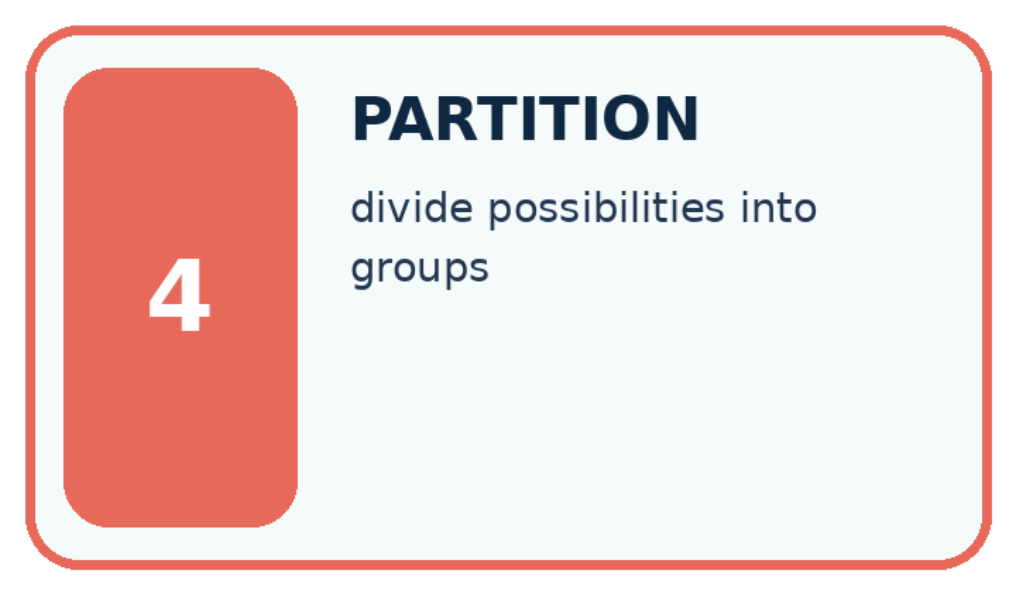
Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.




2
NODE
a question or result point


3
LEAF
final output



Word	My own example	Not this / boundary
decision tree	A sequence of branching questions that routes an input to an output at a leaf.	
node	A point in a graph or tree representing a state, question or result.	
leaf	A final node of a decision tree that supplies an output rather than another question.	
partition	One separated portion of a dataset, such as training, validation or test data.	

Words used in the closer explanation

Word	Meaning
Root	The first question asked by a tree
Branch	A path selected by the answer to a question
Depth	The largest number of question steps from root to leaf

STUDENT LESSON · 6 OF 18

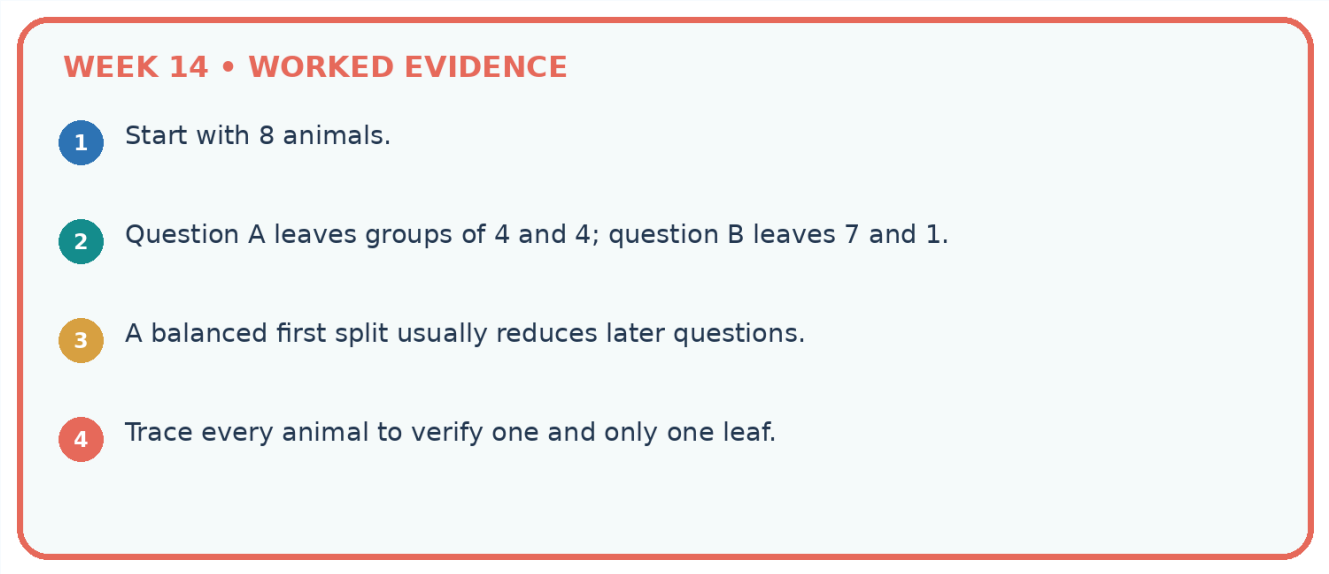
Follow a worked example

> Follow the worked trace for Decision Trees, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. Start with 8 animals.

2. Question A leaves groups of 4 and 4; question B leaves 7 and 1.

3. A balanced first split usually reduces later questions.

4. Trace every animal to verify one and only one **leaf**.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Two paths through the learned tree

ROOT QUESTION	BRANCH ANSWER	LEARNED LEAF
Is near equal to 1	No near is 0	Predict clear
Is near equal to 1	Yes near is 1	Predict blocked

Both paths start at the same root. Tall is not used by this fitted tree; the leaf labels came from training counts.

Worked explanation · Decision Trees

Validation rows are (near=1,tall=1,blocked), (0,0,clear), (0,1,blocked), (1,0,clear). Follow only near: **prediction**s are blocked, clear, clear, blocked. Two of four match.

The **training**-majority baseline predicts blocked, since **training** has four blocked and two clear; it also matches two of four validation labels.

Which validation rows does the tree miss, and which does the baseline miss? Explain why the tall value cannot change this tree's prediction.

Keep the fitted **root** and leaves unchanged while scoring. Equal accuracy does not mean identical errors, and five correct **training prediction**s out of six is not the validation result.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Decision Trees

A fictional toy classifier predicts whether a book needs an illustration label.

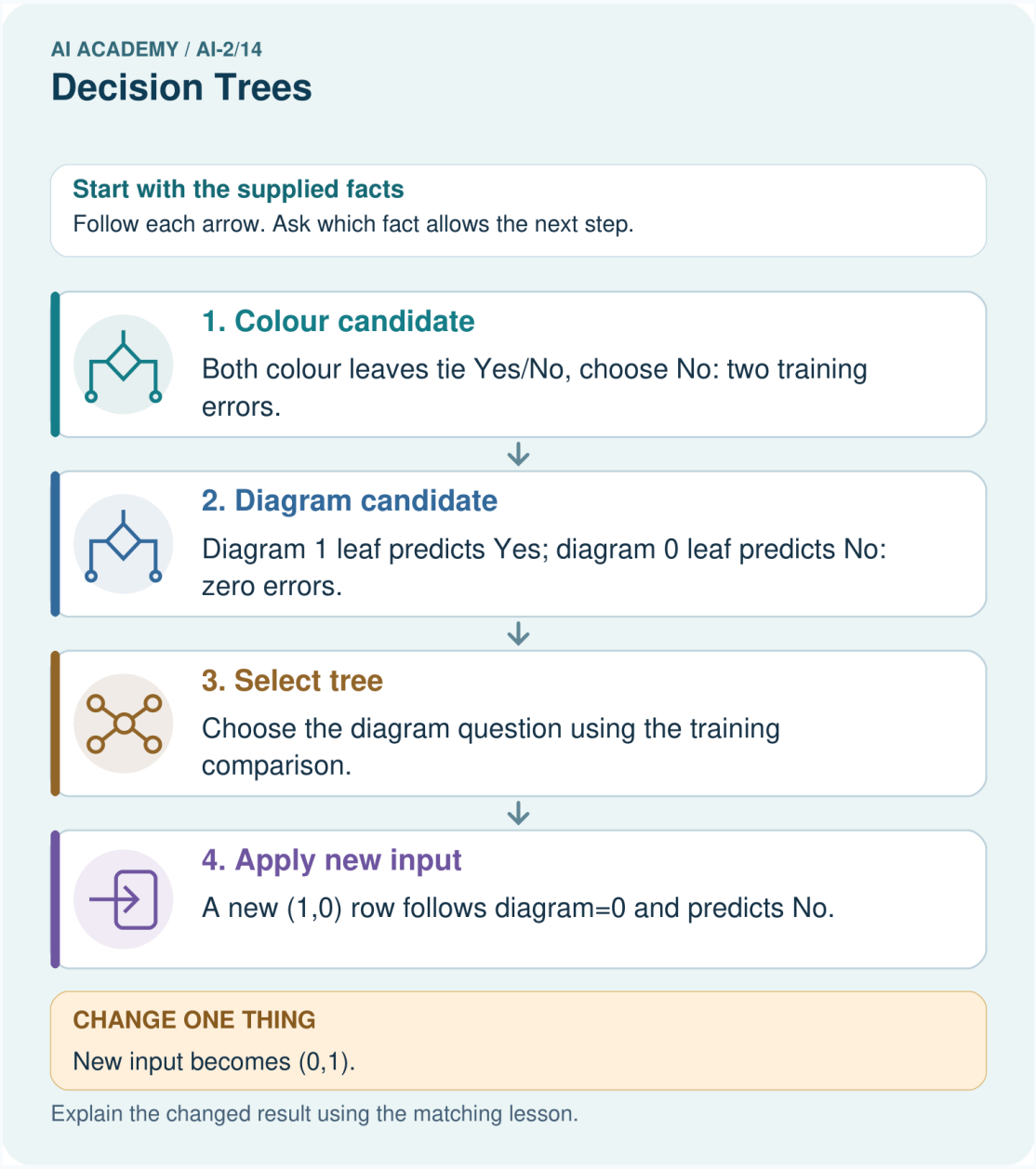
The facts we will use

Supplied facts

Training rows (colour_cover,diagram_inside,label): (1,1,Yes),(1,0,No),(0,1,Yes),(0,0,No).

Supplied facts

Candidates ask colour_cover or diagram_inside; each **leaf** uses its **training** majority. A **leaf** tie chooses No. Select fewer **training** errors.



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Colour candidate	Both colour leaves tie Yes/No, choose No: two training errors.
2. Diagram candidate	Diagram 1 leaf predicts Yes; diagram 0 leaf predicts No: zero errors.
3. Select tree	Choose the diagram question using the training comparison.
4. Apply new input	A new (1,0) row follows diagram=0 and predicts No.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A question becomes a fitted tree only after its leaves and selection procedure are specified. The **training** comparison prefers the feature that separates these labels.

Prediction follows the selected question; it does not reselect the tree using a new case's answer. Save candidate leaves and errors so the choice can be repeated.

What this example does not establish

Keep the stated limit

Zero errors on four toy examples do not establish that diagrams always determine a real book's classification.

Keep the stated limit

Read the labelled picture: Why does changing colour alone not change this tree's **prediction**?

Keep the stated limit

Change one thing: New **input** becomes (0,1). Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Decision Trees claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: A decision tree learns because arrows are drawn between boxes.
Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Record the two candidate questions, **branch** label counts, total errors, tie rules, **depth** limit, stored tree and validation **prediction**s.

Retain all rows, including the blocked example in the near=0 **branch**; removing a disagreement would change the **evidence**.

Check the claim against the facts

Our tree uses a single feature because that question scored better under this procedure. This neither establishes that near causes blockage nor that tall is useless in every dataset.

Next week a nearest-neighbour method will predict by comparing examples instead of following learned questions.

A learner says the tree is reliable because near=1 has only blocked training labels. Rewrite this claim using the leaf evidence, validation score and one limitation.

STUDENT LESSON · 8 OF 18

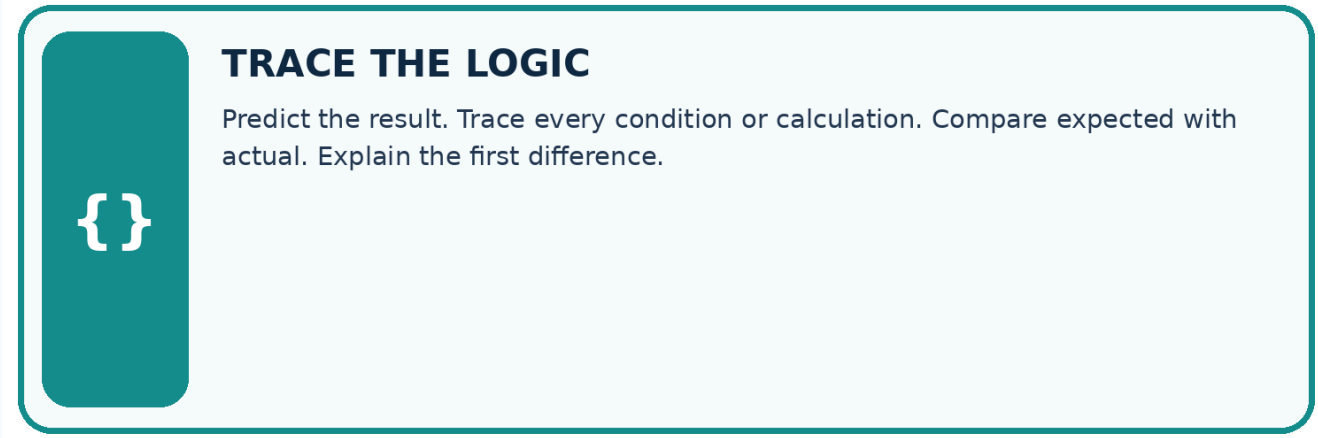
Practise together

> Practise Decision Trees with support, then explain the deciding step in your own words.

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Do not assume near=1 must always predict blocked. If a new **training** set gives that **branch** a clear majority, this same learner will store clear there. A feature name is not the answer.

Use the **training** labels belonging to that **branch**.

Adding more questions can fit finer **training** details but does not guarantee better new-case performance. Record any **depth** choice made during development, and keep final test data separate.

A **leaf** majority describes this **training** group; it is not a promise about every future example.

Practise with less help: Verify the candidate scores

Use workbook W2 for the connected example “**Fitting** a **decision tree**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Fitting** a **decision tree**. Proposed explanation: “A near **branch** must predict blocked”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Why does changing colour alone not change this tree’s **prediction**?

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

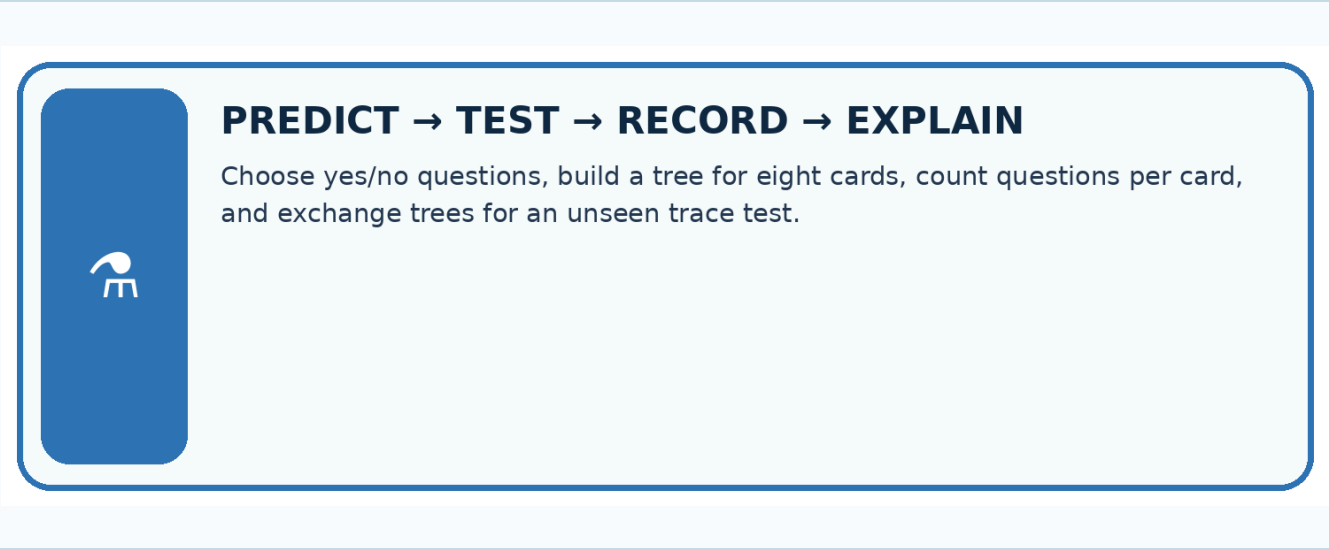
Run the investigation

> Investigate Decision Trees. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Choose yes/no questions, build a tree for eight cards, count questions per card, and exchange trees for an unseen trace test.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

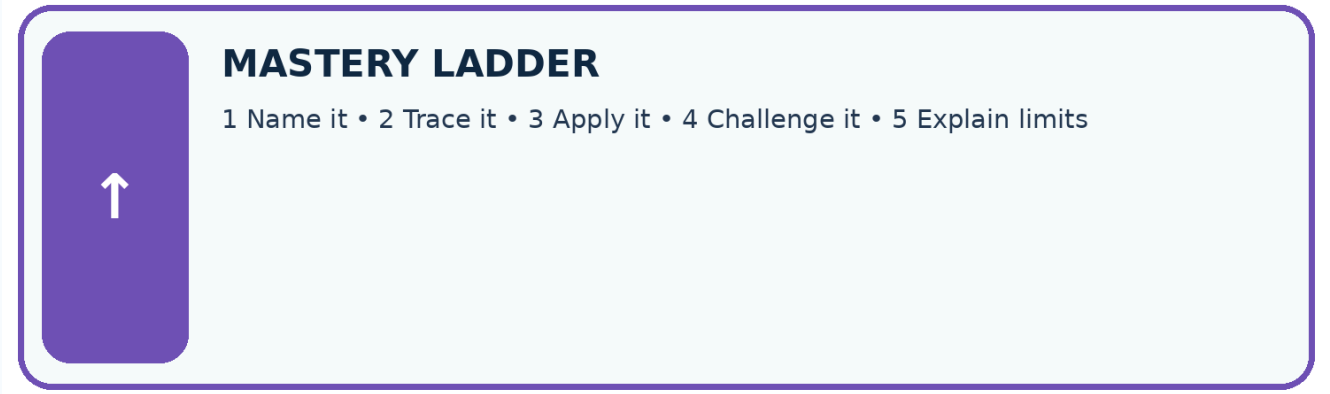
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Decision Trees. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, **branch**, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

> **Write or draw your reasoning and evidence.**

STUDENT LESSON · 11 OF 18

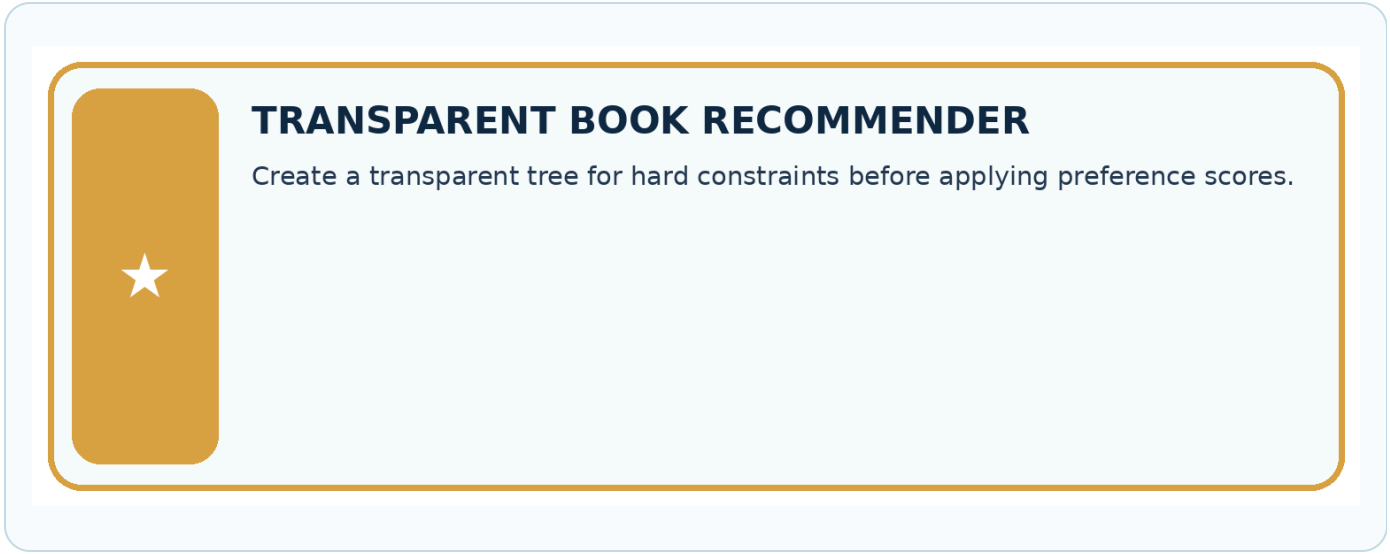
Connect your project

> Apply this week's idea to Transparent Book Recommender. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT Create a transparent tree for hard constraints before applying preference scores.

Trace the system

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Fit or hand-specify recommender classification rules transparently, retaining candidate definitions and **training**-only selection **evidence**.

STUDENT LESSON · 12 OF 18

Show what you know

Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.

✓

INDEPENDENT PROOF

Build a tree that identifies eight mystery animals.

Try a more demanding case

CHALLENGE Build a tree that identifies eight mystery animals.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Decision Trees case for supported practice. Repeated examples do not become fresh assessments.

Week 14 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

First ask animal book? If no, Other. If yes, ask pages<100; yes gives Short Animal, no gives Longer Animal.

The question

Trace 90-page animal and science books.

Worked explanation

The animal book reaches Short Animal; science reaches Other after the first question.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

A decision tree can stop early

Evidence or stage	Value or result
Not an animal book	Other
Animal and fewer than 100 pages	Short Animal
Animal and at least 100 pages	Longer Animal

First ask animal book? If no, Other. If yes, ask pages<100; yes gives Short Animal, no gives Longer Animal.

Trace 90-page animal and science books.

The animal book reaches Short Animal; science reaches Other after the first question.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Fit a new tree independently

New **training** rows (near,tall,label) are (1,0,clear),(1,1,clear),(1,0,clear),(0,1,blocked),(0,0,blocked),(0,1,clear). Use the same candidate order and tie policies. Give both candidates' leaves and errors, then select the tree.

Validation rows are (1,1,clear),(0,0,blocked),(0,1,clear),(1,0,blocked). List tree **prediction**s and compare accuracy with the **training**-majority baseline.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 14 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A tree asks topic is space, then pages at most 60. A space book has 70 pages. Trace both answers and explain why you cannot stop at the first question.

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Create a transparent tree for hard constraints before applying preference scores.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

> Build a tree for science books: first ask science?; then, for yes, pages<=100? Trace science/100 and history/70. What should happen to science with missing pages?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Decision Trees, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional toy classifier predicts whether a book needs an illustration label.

By the end, I can	How I will show it
Fit a root question and leaf predictions from training examples	Explain the deciding step, show a trace or test, and state one limit.
Trace a new example through the learned tree	Explain the deciding step, show a trace or test, and state one limit.

By the end, I can	How I will show it
Compare a shallow tree with a baseline and state its limits	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Decision Trees without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Fit a **root** question and **leaf prediction**s from **training** examples

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional toy classifier predicts whether a book needs an illustration label.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? A new (1,0) row follows diagram=0 and predicts No.

1. Select tree

2. Diagram candidate

3. Colour candidate

4. Apply new **input**

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Apply new **input** A new (1,0) row follows diagram=0 and predicts No. A question becomes a fitted tree only after its leaves and selection procedure are specified.

The **training** comparison prefers the feature that separates these labels. **Prediction** follows the selected question; it does not reselect the tree using a new case's answer.

Save candidate leaves and errors so the choice can be repeated.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: New **input** becomes (0,1). Which conclusion is supported by the supplied facts?

1. The fitted tree predicts Yes.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The fitted tree predicts Yes. It follows diagram=1; cover colour is not the selected question.

Repair a missing building block

Review Types of Classification Error (ai-2/13). Goal: Build a labelled confusion matrix from matched reference and **prediction** pairs. Short practice: Which error causes a sound book to be sent for unnecessary repair review?.

Return to this week and explain what you changed.

Review **Training** and Testing (ai-2/11). Goal: Choose a split unit that matches the intended **prediction** claim. Short practice: How many distinct books are in each repaired role, and why report that as well as scan counts?.

Return to this week and explain what you changed.

Review Operational Labels (ai-2/8). Goal: Apply a category definition using explicit **evidence** and boundary rules. Short practice: Why can we not resolve the disagreement merely by taking a majority of reviewers?.

Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

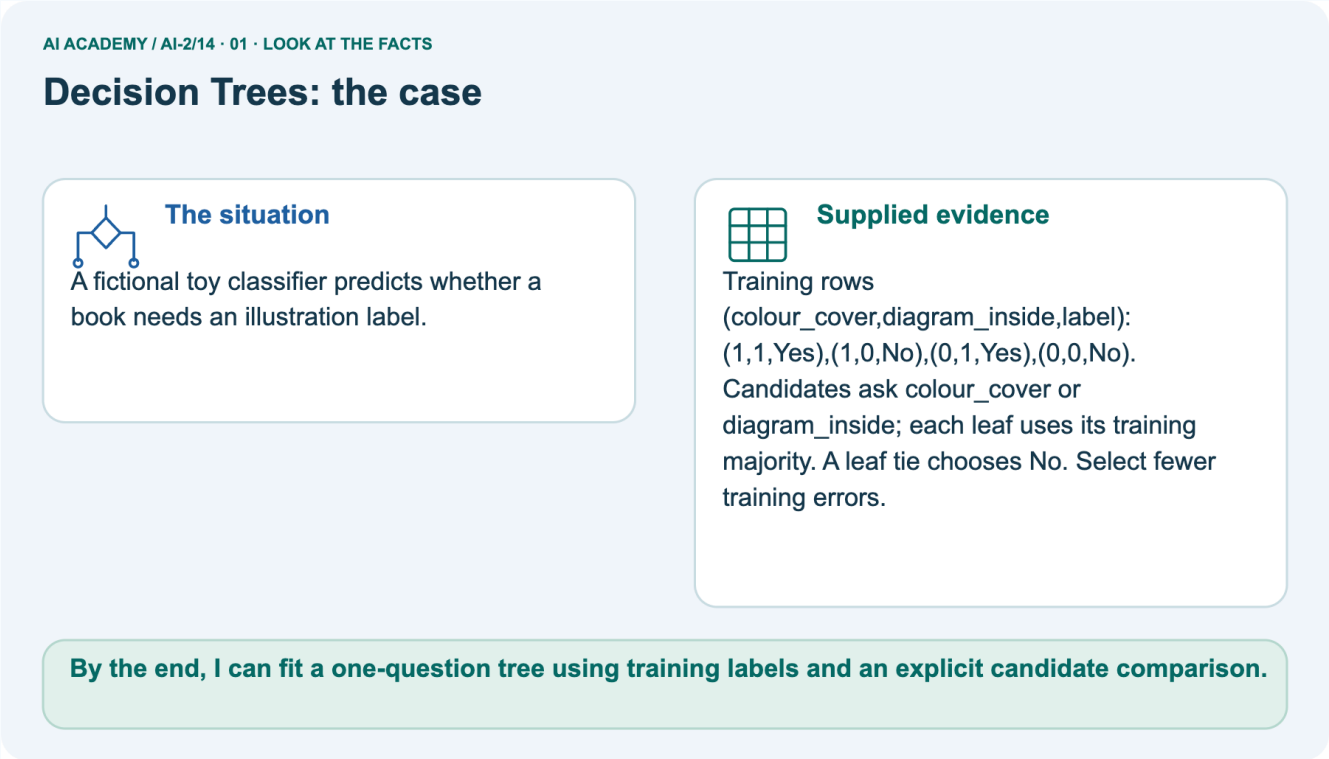
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Decision Trees graphic using its labels, deciding evidence and relevant limitation.

01. Decision Trees: the case



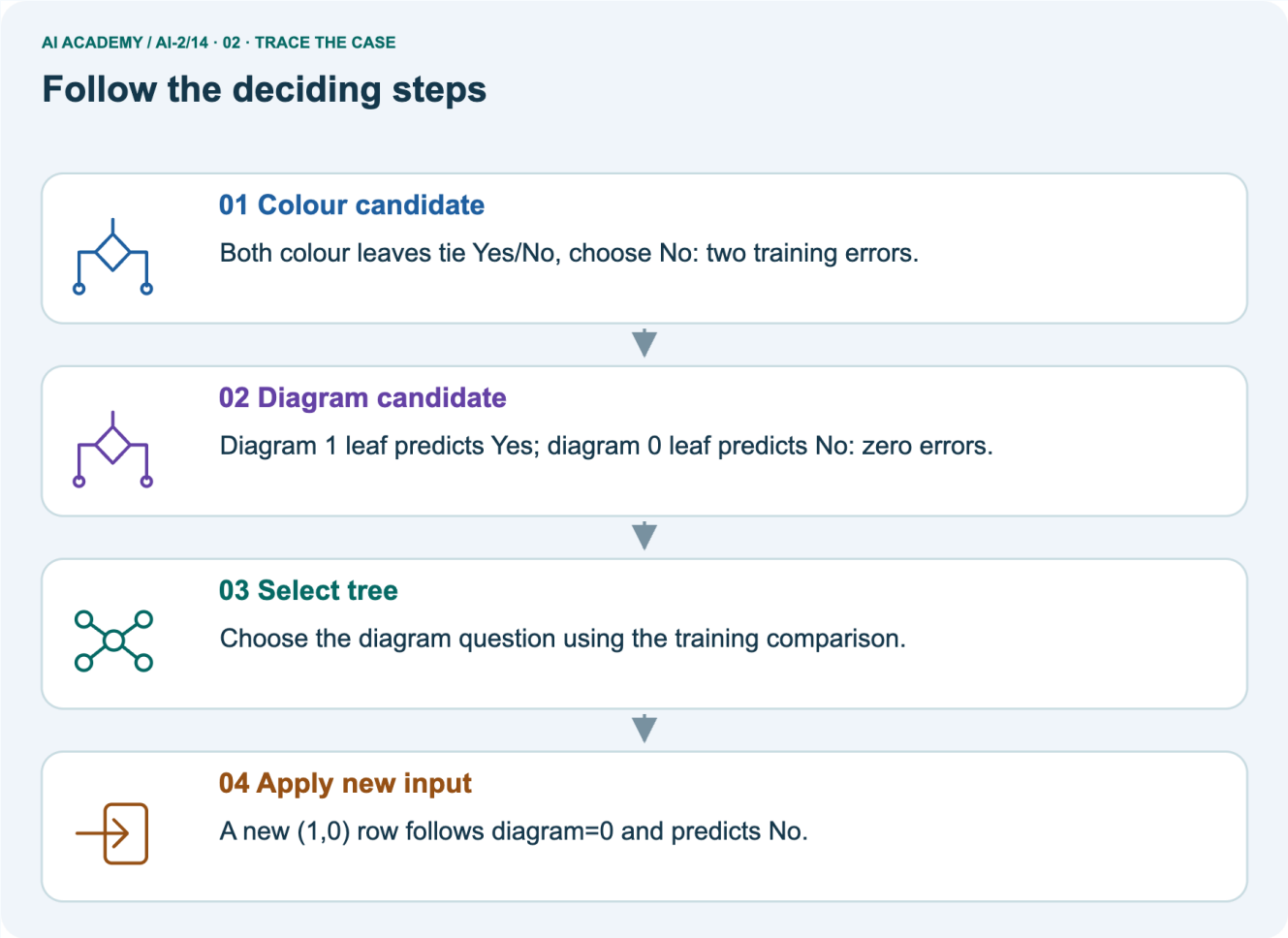
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional toy classifier predicts whether a book needs an illustration label. Supplied facts: **Training** rows (colour_cover,diagram_inside,label): (1,1,Yes),(1,0,No),(0,1,Yes),(0,0,No).

Candidates ask colour_cover or diagram_inside; each **leaf** uses its **training** majority. A **leaf** tie chooses No. Select fewer **training** errors.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Colour candidate: Both colour leaves tie Yes/No, choose No: two **training** errors. → Diagram candidate: Diagram 1 **leaf** predicts Yes; diagram 0 **leaf** predicts No: zero errors. → Select tree: Choose the diagram question using the **training** comparison. → Apply new **input**: A new (1,0) row follows diagram=0 and predicts No.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Two paths through the learned tree

AI ACADEMY / AI-2/14 · 03 · INSPECT THE LESSON MODEL

Two paths through the learned tree

Root question	Branch answer	Learned leaf
Is near equal to 1	No near is 0	Predict clear
Is near equal to 1	Yes near is 1	Predict blocked

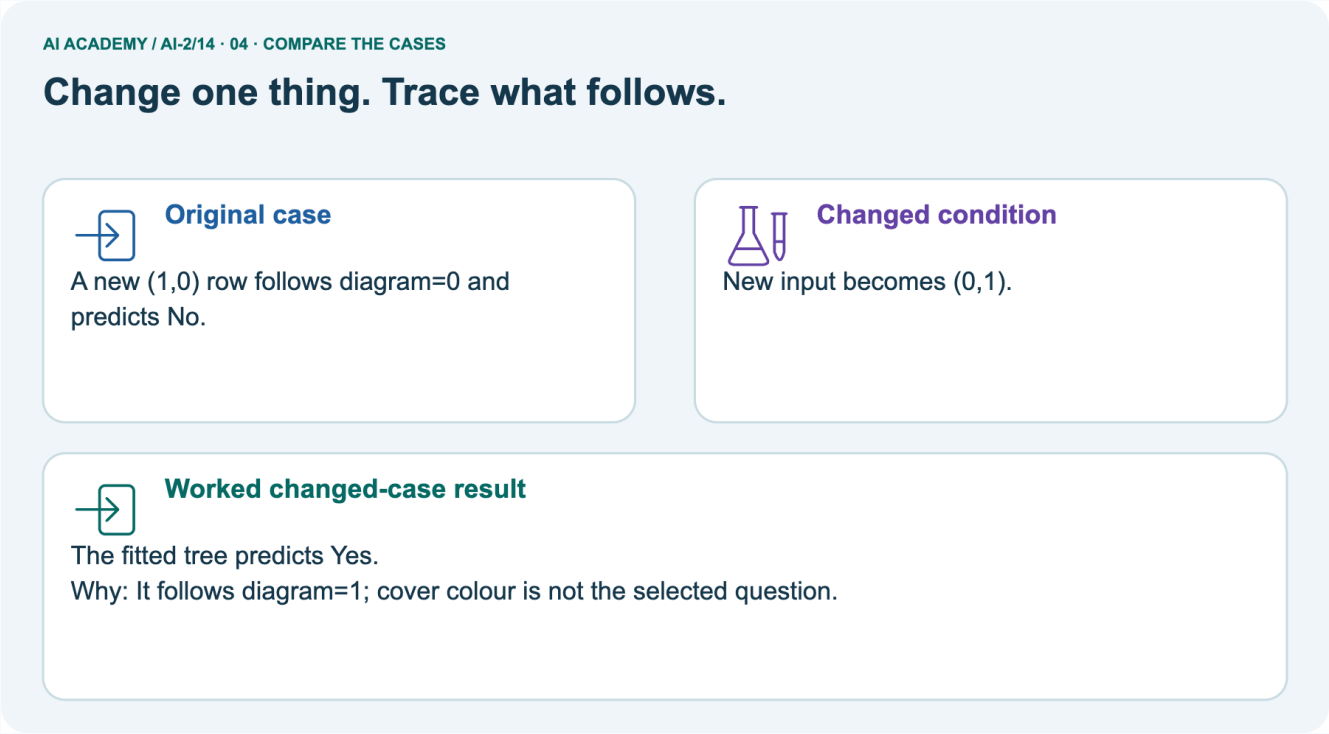
Both paths start at the same root. Tall is not used by this fitted tree; the leaf labels came from training counts.

Follow the labelled picture with the matching explanation. Open the vector diagram

Two paths through the learned tree. Is near equal to 1 | No near is 0 | Predict clear. Is near equal to 1 | Yes near is 1 | Predict blocked

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



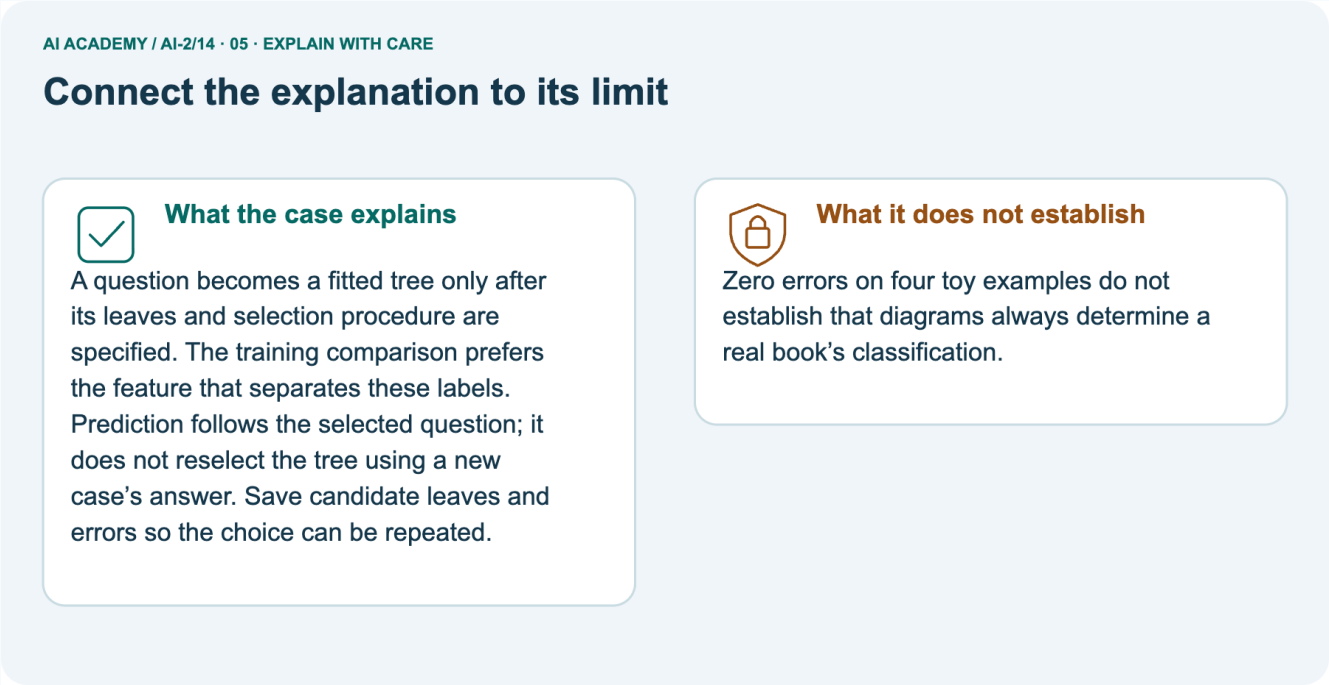
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: A new (1,0) row follows diagram=0 and predicts No. Changed condition: New **input** becomes (0,1). Expected result: The fitted tree predicts Yes.

Reason: It follows diagram=1; cover colour is not the selected question.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

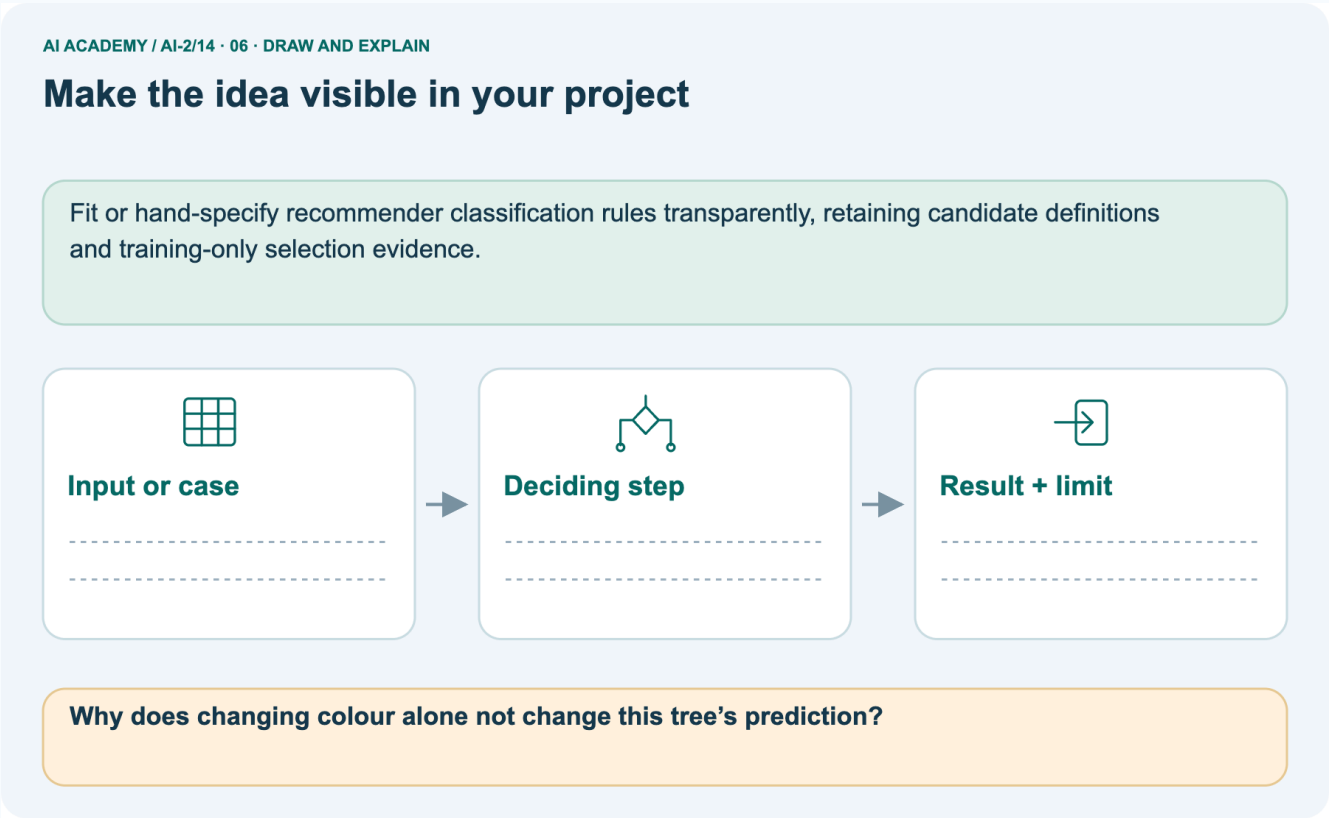
A question becomes a fitted tree only after its leaves and selection procedure are specified. The **training** comparison prefers the feature that separates these labels.

Prediction follows the selected question; it does not reselect the tree using a new case's answer. Save candidate leaves and errors so the choice can be repeated.

Limit: Zero errors on four toy examples do not establish that diagrams always determine a real book's classification.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Fit or hand-specify recommender classification rules transparently, retaining candidate definitions and **training**-only selection **evidence**. Draw your **input**, deciding step, result and limit.

Supported question: Why does changing colour alone not change this tree's **prediction**?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.